CCNA COMMANDS

✅ 1. Basic Cisco IOS Navigation

Task	Command
Enter privileged EXEC mode	`enable`
Enter global config	`configure terminal`
Exit to previous mode	`exit`
Exit to privileged EXEC	`end`
Save configuration	`copy running-config startup-config`
Show running config	`show running-config`
Show startup config	`show startup-config`
View IOS version	`show version`
View IP interfaces	`show ip interface brief`
View protocols	`show protocols`

✅ 2. Interface Configuration

| Task | Command |

Assign IP to interface | `interface fastEthernet 0/0` → `ip address <IP> <MASK>`

Enable interface | `no shutdown`

Disable interface | `shutdown`

Set duplex & speed | `duplex full` / `speed 100`

Show interfaces | `show interfaces`

Show status | `show interfaces status`

✅ 3. VLAN & Trunking (Switching)

| Task | Command |

Show VLANs | `show vlan brief`

Create VLAN | `vlan <ID>`

Name VLAN | `name <name>`

Assign port to VLAN | `interface fa0/1` → `switchport mode access` → `switchport access vlan <ID>`

Configure trunk | `interface fa0/24` → `switchport mode trunk` → `switchport trunk allowed vlan <list>`

Set native VLAN | `switchport trunk native vlan <ID>`

Show trunk info | `show interfaces trunk`

✅ 4. STP (Spanning Tree Protocol)

| Task | Command |

View STP | `show spanning-tree`

Set bridge priority | `spanning-tree vlan <ID> priority <value>`

Make root bridge | `spanning-tree vlan <ID> root primary`

Portfast | `spanning-tree portfast`

BPDU Guard | `spanning-tree bpduguard enable`

Change STP mode | `spanning-tree mode rapid-pvst`

✅ 5. Inter-VLAN Routing

| Task | Command |

Enable routing on L3 switch | `ip routing`

Create SVI | `interface vlan <ID>` → `ip address <IP> <MASK>` → `no shutdown`

Router-on-a-stick subinterface | `interface g0/0.<ID>` → `encapsulation dot1q <ID>` → `ip address <IP> <MASK>`

✅ 6. Static Routing

| Task | Command |

Add static route | `ip route <network> <mask> <next-hop>`

Default route | `ip route 0.0.0.0 0.0.0.0 <next-hop>`

View routes | `show ip route`

✅ 7. RIP

Enable RIP | `router rip`

Specify networks | `network <network>`

Enable Version 2 | `version 2`

Disable auto-summarization | `no auto-summary`

View RIP routes | `show ip route rip`

✅ 8. OSPF

Enable OSPF | `router ospf <process-id>`

Advertise networks | `network <ip> <wildcard> area <area>`

Set router ID | `router-id <ID>`

Show OSPF neighbors | `show ip ospf neighbor`

Show OSPF routes | `show ip route ospf`

Show OSPF status | `show ip ospf`

✅ 9. EIGRP

Enable | `router eigrp <AS>`

Advertise networks | `network <network>`

Disable auto-summary | `no auto-summary`

Show neighbors | `show ip eigrp neighbors`

Show routes | `show ip route eigrp`

✅ 10. DHCP

Enable DHCP service | `service dhcp`

Create pool | `ip dhcp pool <name>`

Set network | `network <IP> <MASK>`

Default gateway | `default-router <gateway>`

Exclude addresses | `ip dhcp excluded-address <start> <end>`

Show leases | `show ip dhcp binding`

✅ 11. NAT / PAT

Inside interface | `int g0/0` → `ip nat inside`

Outside interface | `int s0/0` → `ip nat outside`

Static NAT | `ip nat inside source static <private-IP> <public-IP>`

Dynamic NAT | `ip nat pool <NAME> <start> <end> netmask <mask>` → `access-list <n> permit <network>` → `ip nat inside source list <n> pool <NAME>`

PAT (overload) | `ip nat inside source list <n> interface <int> overload`

Verify NAT | `show ip nat translations`

✅ 12. ACL (Access Control Lists)

Standard ACL

Create ACL | `access-list <number> permit/deny <IP> <wildcard>`

Apply to interface | `ip access-group <number> in/out`

Extended ACL

`access-list 100 permit tcp 192.168.1.0 0.0.0.255 any eq 80`

Show ACLs | `show access-lists`

✅ 13. IPv6

Enable IPv6 | `ipv6 unicast-routing`

Set IPv6 on interface | `interface g0/0` → `ipv6 address <address>/<prefix>`

Static route | `ipv6 route <network> <next-hop>`

Show IPv6 | `show ipv6 interface brief` / `show ipv6 route`

✅ 14. Wireless (Basic)

Set SSID | `ssid <name>`

Set WPA2 PSK | `wpa2-psk <password>`

Show wireless | `show wireless ssid summary`

✅ 15. WAN / PPP / HDLC

PPP encapsulation | `interface s0/0` → `encapsulation ppp`

HDLC default | `encapsulation hdlc`

PPP PAP | `ppp pap sent-username <username> <password>`

PPP CHAP | `ppp chap hostname <name>` → `ppp chap password <pass>`

✅ 16. Security / Encryption

Enable passwords:

```
enable secret <password>
line vty 0 4
 password <password>
 login
```

Encrypt passwords | `service password-encryption`

Banner | `banner motd # unauthorized access prohibited #`

✅ 17. Device Management

Hostname | `hostname <name>`

Set domain | `ip domain-name <domain>`

Generate SSH key | `crypto key generate rsa`

Enable SSH | `ip ssh version 2`

User account | `username <user> privilege 15 secret <pass>`

Show log | `show logging`

✅ 18. Backup & Restore

Copy config to TFTP | `copy running-config tftp`

Copy back | `copy tftp running-config`

Erase startup config | `erase startup-config`

Reload device | `reload`

✅ 19. Troubleshooting

Ping | `ping <IP>`

Traceroute | `traceroute <IP>`

CDP neighbors | `show cdp neighbors detail`

Check ARP | `show ip arp`

Show interfaces errors | `show interfaces`

DNS test | `nslookup <host>`

✅ 20. Packet Capture & Debug

Debug IP packets | `debug ip packet`

Stop debug | `undebug all` or `u all`

Show logs | `show logging`

What is a Wildcard Mask?

A **wildcard mask** tells the router **which bits to check and which bits to ignore** in an IP address.

✅ **0 = Match this bit exactly**

✅ **1 = Ignore this bit (don't care)**

✅ Example 1: Exact match of a single host

IP: `192.168.1.10`

Wildcard: `0.0.0.0`

Means: **match exactly this host**

ACL example:

```
access-list 10 permit 192.168.1.10 0.0.0.0
```

✅ Example 2: Match entire network

Network: `192.168.1.0`

Subnet mask: `255.255.255.0`

Wildcard: `0.0.0.255`

Because 255.255.255.0 → subtract from 255.255.255.255:

```
255.255.255.255
- 255.255.255.0
= 0.0.0.255
```

ACL:

```
access-list 10 permit 192.168.1.0 0.0.0.255
```

✔️ Matches all hosts from `.1` to `.254`

✅ Quick trick to convert

Wildcard = 255.255.255.255 – Subnet Mask

Example:

Subnet mask: `255.255.254.0`

```
255.255.255.255
-255.255.254.0
=0.0.1.255
```

✅ Example 3: Match only specific range

Match only first 16 hosts of 192.168.10.0:

IPs: `192.168.10.0 – 192.168.10.15`

Wildcard:

```
0.0.0.15      (because 15 = 0000 1111)
```

ACL:

```
access-list 10 permit 192.168.10.0 0.0.0.15
```

✅ Where Wildcards are used?

Feature	Example
ACLs	`access-list 10 permit 192.168.1.0 0.0.0.255`
OSPF Network statement	`network 10.1.0.0 0.0.255.255 area 0`
EIGRP	`network 172.16.0.0 0.0.3.255`

✅ Wildcard Summary Table

Wildcard	Meaning
`0.0.0.0`	Match exactly 1 IP
`0.0.0.255`	Match 256 addresses in last octet
`0.0.255.255`	Match full Class B range
`255.255.255.255`	Match ANY IP

✅ One-line definition:

✔️ **Wildcard mask tells which parts of IP to match and which to ignore — 0 = match, 1 = ignore.**